

Boilertronics

Project Backlog

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Problem Statement:

Many games are made just for you to relax and have fun. But, as computer science students, our fun is through solving problems while taking into account many factors. In this game you must optimize a factory for multiple different metrics such as time efficiency, total cost, and profit per cycle. Taking inspiration from games made by Zachtronics, but with our own programming language, mechanics, and story. Our main goal with this game is to allow for people to both relax and have fun while staying sharp and keeping their brain engaged.

Background Information:

Target Users:

Boilertronics is a game created for anyone who wants to have fun while improving their problem solving and programming skills. The majority of games we see today mainly focus on the enjoyment factor of the game, and the pleasant nature of the user's experience. We, however, are designing a unique, challenging puzzle experience that provides engaging entertainment for both inexperienced and competent programmers. Ideally, as an accessible puzzle game, this game should be "easy to pick up, but difficult to master" – a philosophy that will be encouraged by a simple 'assembly-like' programming language, and a small selection of interactable components complemented by a rigorous scoring system. Compared to the average strategy game, Boilertronics is designed to encourage extensive optimization throughout a user's solution, making the user compete against themselves to progress as a programmer and problem solver.

Similar Games:

A game developer known as "Zachtronics" is well known for their specific style of puzzle games that focus on unique mechanics that often integrates programming mechanisms into their games. Most of these games mirror real programming and electronic principles, often in a simplified or reduced form. Since 2011, their games have had notable influence in the puzzle game genre – thus creating the "Zach-like" puzzle sub-genre.

Boilertronics will identify as a "Zach-like", as its focus on programming and optimization closely mirrors its peers in this subgenre. However, we will aim to differentiate ourselves by creating a familiar, yet unique puzzle experience with an interactable, isometric, three-layered "factory" world with programmable "claw" and "rotator" mechanisms.

Limitations:

The current market lacks any “Zach-likes” that fully commit to building a factory, let alone any that also manage multiple layers of placeable objects controlled with a custom programming language. These elements are spread across the entire “Zach-like” subgenre of puzzle games, but no single game has everything that we wish to include. We feel that these elements will lead to interesting puzzles with complex, optimizable solutions. This is a recipe for a great puzzle game and an excellent “Zach-like.” The others don’t have ‘limitations’ per se, but they don’t have all the features that we are planning to include.

Functional Requirements:

- 1.) As a user, I want there to be movable, placeable tiles to put on the layers.
- 2.) As a developer, I want each “step” to allow the programmable objects to move.
- 3.) As a developer, I want the view of the game/tiles to be isometric.
- 4.) As a user, I would like an overarching story.
- 5.) As a developer, I want to have a level creator framework ensuring additional levels can be created, with an interface to create levels.
- 6.) As a user, I would like a leaderboard system to compare myself to my previous scores.

- Level Functionality

- 7.) As a developer, I want to make sure that users can only place tile objects where and when permitted.
- 8.) As a user, I want to be able to configure the time each ‘step’ takes in running my solution to the level (speed up / slow down / pause).
- 9.) As a user, I want to stop the level if an error or collision occurs in the user solution.
- 10.) As a user, I want to be able to run my solution step by step.
- 11.) As a user, I want to see when my solution encounters an error.
- 12.) As a developer, I want to require the user to reset their solution before being able to change their solution.
- 13.) As a user, I want to be able to reset my solution at any point while I’m stepping through my solution.
- 14.) As a user, I want to be able to save and load my solutions.
- 15.) As a developer, I would like a mechanism that delineates the “quality” of different interactable components in the level that have varying cost/performance factors based on their operation cost.
- 16.) As a user, I would like to be graded on my solution’s statistics such as time to finish, total resources consumed, resources produced per second, and other such factors.
- 17.) As a user, I would like to be given grading “cutoffs” to have a point of reference for ideal solutions.
- 18.) As a user, I would like to be presented with unchangeable obstructions, including (but not limited to) such obstructions such as missing tiles on any layer.

- **Programming Language/Editor**

- 19.) As a user, I want to have multiple sidebar terminal windows where I can write my code for the level.
- 20.) As a user, I would like to be able to write code in the in-game code editor.
- 21.) As a user, I want a visual indicator that the code I have written is either correct or has some form of syntax error.
- 22.) As a developer, I want the programming language to support stoppable commands like "grab," "wait," "move."
- 23.) As a developer, I would like the programming language to interact with defined, scriptable objects to interact with the level in an iterative, step based process.
- 24.) As a developer, I would like the validity of the script to be checked on runtime, on a line by line basis.
- 25.) As a developer, I would like there to be dynamic error checking in the code terminal, if time allows.
- 26.) As a user, I want the code editor to have different "tabs" corresponding to different elements within the level.
- 27.) As a developer, I would like to further extend the instruction set to support more registers and the concept of "null" if time allows.
- 28.) As a user, I would like there to be a technical manual on the in game programming language.
- 29.) As a developer, I would like there to be a gradual reveal of information pertaining to the full range and scope of the programming language to the user over the course of level progression.
- 30.) As a user, I would like to have the scripts run synchronously such that all the steps happen at the same time.

- **UI/UX**

- 31.) As a user, I want to be able to navigate to a level selector via a main menu.
- 32.) As a user, I want to be able to navigate between levels via the level selector.
- 33.) As a user, I want to be able to navigate between different groups of levels via the level selector.
- 34.) As a user, I would like a quick reference panel / menu for global hints based on progress.
- 35.) As a user, I would like a hint system that would allow the user to see the quick reference for the concept introduced in the level.
- 36.) As a user, I want the level UI to be divided into different sections (Statistics, Placeable Items, Exit Menu, Time Interaction [Pause, play], etc).
- 37.) As a user, I want to be able to move and re-order my code editor tabs.
- 38.) As a user, I want the placeable elements on the floor layer to have visuals, as well as for the floor layer to have default visuals.
- 39.) As a user, I want the placeable elements on the movable layer to have visuals.
- 40.) As a user, I want placeable elements on the claw layer to have visuals.

- 41.) As a user, I want to be able to separate between the different layers in an isometric view.
- 42.) As a user, I want to have the placement grid be aligned with the layer that the user is interacting with.
- 43.) As a user, I want to toggle the view of a specific layer on and off.
- 44.) As a user, as I am placing tiles, I want the tiles on the corresponding layer to be more opaque than others for visibility improvements.
- 45.) As a user, I want some sprites to be animated.
- 46.) As a developer, I want trains to connect the levels in the level select menu.
- 47.) As a user, I would like sound effects to accompany actions from interactable game mechanisms, time iteration, and other mechanics.
- 48.) As a user, I want to be able to change the game's volume settings.

- Server Integration

- 49.) If time allows, as a user, I want to be able to create and login to an account.
- 50.) If time allows, as a user, I want to be able to play the game on Steam
- 51.) If time allows, as a user, I would like to be able to see other people's scores online and compare my score to theirs.
- 52.) If time allows, as a user, I would like to view the score my friends got online and compare my scores to theirs.
- 53.) If time allows, as a user, I would like to design my own levels and share them online.

- Extras

- 54.) As a user, I would like an achievements system derived from game and challenge completion.
- 55.) As a developer, I want a level editing system to make designing new levels easier.

Non-functional Requirements:

- 56.) As a user, I want good-looking tile sprites.
- 57.) As a user, I want responsive and intuitive buttons.
- 58.) As a user, I would like the game's visuals to be appealing and easy to read.
- 59.) As a user, I want to be able to scale the UI to make it easier to read.
- 60.) If time allows, as a user, I would like to have colorblind modes.
- 61.) If time allows, as a developer, I would like to include easter eggs into the game.
- 62.) If time allows, as a user, I would like a block mode for the scripts rather than a strictly terminal based approach.
- 63.) If time allows, as a user, I would like to have a modding framework.
- 64.) If time allows, as a user, I would like more levels with different gimmicks such as pipes, electricity, heat, and other similar mechanics.